



Genetic/Hereditary Factor as Cause of Cancer



Main Examples of Genetic Causes

A. Autosomal Dominant Hereditary

- BRCA1/2
- APC
- MEN1/2
- TP53

B. Autosomal Recessive Hereditary

- MUTYH
- ATM
- FANC
- XPA & XPC

C. Oncogenes

- HER2
- MYC & NMYC
- EGFR & KRAS



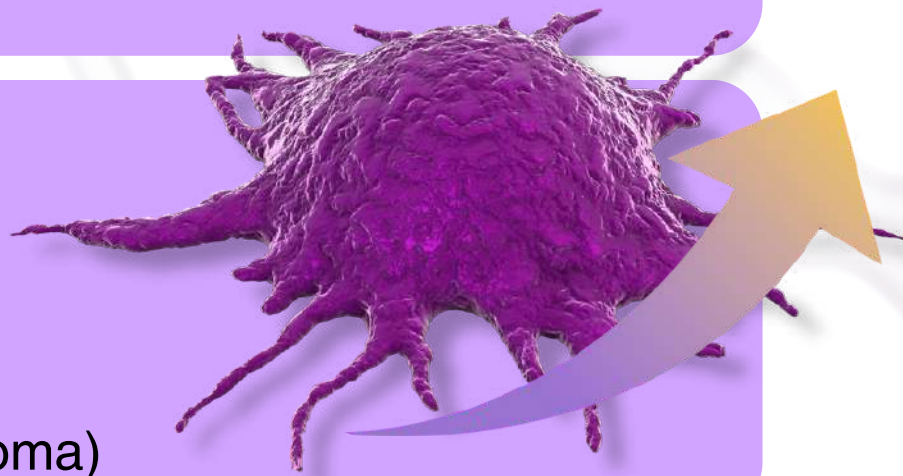
Types of Cancer Linked to Genetic Mutations

Autosomal Dominant Hereditary:

- BRCA1 and BRCA2 - breast and ovarian (also ↑ risk for prostate, colorectal, and pancreatic cancer)
- MEN1 and MEN2 - endocrine tumors
- APC - multiple colorectal and upper GI adenomatous polyps
- TP53 - breast cancer, sarcomas, leukemia, and brain tumors

Autosomal Recessive Hereditary:

- MUTYH - Colorectal cancer, colonic adenoma
- ATM - Non-Hodgkin lymphoma, leukemia
- FANC - Leukemia (increased risk for Brain tumors & Wilms tumor)
- XPA & XPC - Skin cancer (Basal cell, squamous cell carcinoma, melanoma)



Contribution and Prevention to Cancer Development

Category	Gene	How it Causes Cancer
Autosomal Dominant Hereditary	BRCA1/2	Loss of DNA repair function
	APC	Failure to control cell growth and division
	MEN 1/2	Overactive endocrine cell growth regulation
	TP53	Loss of tumor suppressor control
Autosomal Recessive Hereditary	MUTYH	Defective base excision repair
	ATM	Impaired DNA damage response
	FANC	Defective DNA crosslink repair
	FANC	Defective nucleotide excision repair
Oncogenes	FANC	Overexpression causing increased cell signaling
	FANC	Overactive transcription factors driving proliferation
	FANC	Constant cell growth signaling

Prevention:

- Genetic Testing & Counseling – Detects inherited mutations early for timely prevention.
- Lifestyle Changes – No smoking, limit alcohol, healthy diet, regular exercise.
- Medications – Lowers risk or prevents recurrence in high-risk groups.
- Risk-Reducing Surgery – Preventive mastectomy or ovary/fallopian tube removal for BRCA carriers.



Real Life Example



Angelina Jolie

Jolie tested positive for **BRCA1**. Her mother died due to ovarian cancer. It was estimated that she had an **87% risk of breast cancer** and a **50% risk of ovarian cancer**. She underwent a **prophylactic double mastectomy** in 2013 and **salphingo-oophorectomy** in 2015 to reduce her risk of ovarian cancer.



Role of Nurses in Prevention and Education

Primary Prevention:

- Health promotion (lifestyle changes)
- Genetic counseling/testing encouragement
- Psychological support

Secondary Prevention:

- Screening promotion (mammogram, colonoscopy)
- Encourage follow-up adherence
- Teach self-screening (BSE, skin checks)

Tertiary Prevention:

- Family education & counseling
- Symptom management & treatment adherence
- Psychosocial support for coping and communication